

Brian Rivera

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Department of Psychology
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EDUCATION

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| 2021 | Ph.D. in Educational Psychology, Educational Neuroscience Concentration
Department of Educational Studies, University of Alabama, Tuscaloosa, AL
Advisor: Firat Soylu, PhD
<i>Dissertation: The neurocognitive mechanisms of arithmetic knowledge verification</i> |
| 2015-2016 | Post-baccalaureate studies, Cognition and Action Lab
University of Massachusetts, Amherst, MA
Advisor: Rebecca Spencer, PhD |
| 2013 | B.A., Scientific and Philosophical Studies of the Mind
Franklin & Marshall College, Lancaster, PA |

ACADEMIC POSITIONS

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| 2023- current | Assistant Research Professor
Department of Psychology, University of Nebraska-Lincoln, Lincoln, NE |
| 2021- 2023 | Postdoctoral Research Associate
Department of Psychology, University of Nebraska-Lincoln, Lincoln, NE
Advisor: Timothy Nelson. PhD |

PUBLICATIONS

Published Papers

- Salehzadeh, R., **Rivera, B.**, Man, K., Jalili, N., & Soylu, F. (2023). EEG Decoding of Finger Numeral Configurations With Machine Learning. *Journal of Numerical Cognition*, 9(1), 206-221.
<https://doi.org/10.5964/jnc.10441>
- Rivera, B.**, & Soylu, F. (2021). Incongruity in fraction verification elicits N270 and P300 ERP effects. *Neuropsychologia*, 161, 108015. <https://doi.org/10.1016/j.neuropsychologia.2021.108015>
- Soylu, F., **Rivera, B.**, Anchan, M., & Nathaniel, N. (2019). ERP differences in processing canonical and non-canonical finger numeral representations. *Neuroscience Letters*.
<https://doi.org/10.1016/j.neulet.2019.04.032>

Conference Proceedings

Rivera, B., & Soyly, F. (2018). Semantic Processing in Fraction Comparison: An ERP Study. Paper submitted to the 40th Annual Conference of the Cognitive Science Society. Madison, WI: Cognitive Science Society

PRESENTATIONS

National and International Conference Presentations

Rivera, B. (2024). What Fraction am I Thinking of? Decoding Fraction Magnitude from EEGs. Poster presented at the Annual Meeting of the Mathematical Cognition and Learning Society. Washington D.C.: June, 2024

Rivera, B. (2022). The Neurocognitive Mechanisms of Arithmetic Verification in Children. Poster presented at the Cognitive Development Society Bi-Ennial Conference. Madison, WI: April, 2022.

Rivera, B., & Soyly, F. (2019). The ERP Effects of Shared Components in Fraction Comparisons. Poster presented at the Annual Meeting of the Mathematical Cognition and Learning Society. Carleton University, Ottawa, Canada: June, 2019

Rivera, B., & Soyly, F. (2018). The Effects of Shared Components in Fraction Comparisons: Evidence From ERPs. Poster presented at the 6th Biennial International Mind Brain and Education Conference. The University of Southern California, Los Angeles, CA: September, 2018

Rivera, B. (2018). Online Experiments with Wolfram Technologies. Poster presented at the 2018 Wolfram Technology Conference. Urbana-Champaign, IL: October, 2018

Rivera, B., Shannon, N., & Soyly, F. (2017). ERP Markers for Number Gesture Processing. Poster presented at SACNAS The National Diversity in STEM Conference. Salt Lake City, UT: October, 2017

Rivera, B. & Park, J. (2016). Chunking During Child Visoumotor Sequence Learning. SPINES Summer Program, Marine Biology Lab, Woods Hole, MA: July, 2016

TEACHING EXPERIENCE

Course Instructor

University of Alabama, College of Education - *Research Methods in Ed. Neuro*, Spring 2021

University of Alabama, College of Education - *Intro to Learning Skills & Strategies*, Spring 2020

University of Alabama, OLLI - *A History of Information*, Summer 2019

Teaching Assistant

University of Alabama, College of Education - *Intro to Learning Skills & Strategies*, Fall 2019

University of Alabama, College of Education - *Research Methods in Ed. Neuro*, Spring 2018

INVITED PRESENTATIONS

Invited Speaker

Rivera, B. (2024). Math Development and Learning Difficulties. Guest lecture for Developmental Psychopathology, Teachers College, Columbia University, New York.

Rivera, B. (2022). How Does the Brain Do Math? Lessons From the Study of Fractions. Talk presented at the Mathematics Department Colloquium series, University of Nebraska, Omaha.

AWARDS AND FELLOWSHIPS

2023	ERP Boot Camp Attendance Award, San Diego, CA
2018	Wolfram Summer School Attendance Award, Waltham, MA